

VAT Relief on Shopify — Test Guide

Two implementations, run side by side, on the same cart. One is Betsy's discount approach. One makes the merchandise itself non-taxable. The point of the exercise is to see what each does to the *tax lines on the order*, not what it does to the checkout page.

Project: `checkout-blocks-lab` · ResMed Shop UK · prepared for Izam, 23 August 2026

Read this first — the one number that decides everything

ResMed UK prices are **VAT-inclusive**. The cart screenshot proves it: subtotal £191.00, total £191.00, and “VAT 20% £31.83” sitting *inside* the £191.00.

So the VAT element of a gross price is **gross × 20/120 = gross ÷ 6 = 16.667%**, not 20%. On a £159.00 mask that is **£26.50**, not £31.80. A flat “20% deduction” over-refunds £5.30 on that one line.

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1. Prerequisites and one-time setup

1.1 What you need

- **A development store with Plus features enabled.** Partner dashboard → Stores → your dev store → edit → enable the Plus / developer preview features. Without this the checkout UI extension will not appear in the checkout editor.
- **Shopify CLI** — `npm install -g @shopify/cli@latest`
- **Node** `>=20.19 <22` or `>=22.12`. The repo sets `engine-strict=true`, so a wrong version fails loudly rather than silently.

1.2 Store settings that must be right before anything else

Get these wrong and both tests produce nonsense that looks plausible.

Setting	Value
Settings → Taxes and duties → United Kingdom	VAT 20% collected
Settings → Taxes and duties → “All prices include tax”	ON — this is the whole basis of the arithmetic
Settings → Markets	United Kingdom active, currency GBP
Shipping	Set one free rate so shipping doesn't muddy the totals

1.3 Test products

Create these two. The prices are the real ones from the live UK site so the numbers in this guide match what you will see.

Product	Price (gross)	Tags	Charge tax
AirTouch N30i – Nasal Mask	£159.00	VAT Exempt Opt	Yes
REMZZZ Mask Liner X32	£32.00	— none —	Yes

The liner is the control. If it ever loses VAT, something is wrong with the eligibility gate.

1.4 Install the app

```
cd ~/Documents/Projects/checkout-blocks-lab
npm install
shopify app config link      # writes client_id into shopify.app.toml
shopify app dev              # press P, install into the dev store
```

1.5 Baseline — do this before either test

Add both products to the cart, go to checkout, and record the numbers. Everything afterwards is measured against this.

Line	Gross	VAT inside it
AirTouch N30i	159.00	26.50
REMZZZ liner	32.00	5.33
Total	191.00	31.83

If your checkout does not show **£191.00 total with £31.83 VAT included**, stop and fix the store settings in 1.2. Do not proceed.

2. Test A — discount function (Betsy's approach)

Lives in `extensions/vat-relief-discount`. It reads the cart line property `_vat_relief`, checks the product actually carries the `VAT Exempt Opt` tag, and applies a fixed-amount product discount equal to that line's VAT element.

2.1 Deploy the function

```
shopify app deploy
```

2.2 Register it as an automatic discount

The function has no admin UI, so you create the discount through the GraphQL API rather than the Discounts page. Use the **GraphiQL app** on the dev store (install “Shopify GraphiQL App”) or the Admin API.

First get the function ID:

```
query {
  shopifyFunctions(first: 25) {
    nodes { id title apiType app { title } }
  }
}
```

Find the node whose `apiType` is `discount` and whose app is your dev app. Then:

```
mutation {
  discountAutomaticAppCreate(automaticAppDiscount: {
    title: "VAT relief (discount test)"
    functionId: "PASTE_FUNCTION_ID_HERE"
    startsAt: "2026-08-23T00:00:00Z"
    discountClasses: [PRODUCT]
  }) {
    automaticAppDiscount { discountId title status }
    userErrors { field message }
  }
}
```

`discountClasses: [PRODUCT]` matters — the function returns early if the product class is not enabled, so an order-class-only discount silently does nothing.

2.3 Put the declaration on a cart line

The function reads `_vat_relief`. You need something to write it. For a quick test you do not need theme code at all.

Fast path — browser console

On the storefront, empty the cart, then paste this. Replace the variant IDs with your own (find them in the product page source or via `/products/handle.js`).

```
// Mask WITH the declaration
await fetch('/cart/add.js', {
  method: 'POST',
  headers: {'Content-Type': 'application/json'},
  body: JSON.stringify({
    items: [{
      id: MASK_VARIANT_ID,
      quantity: 1,
      properties: {
        '_vat_relief': 'true',
        '_vat_relief_declared_at': new Date().toISOString()
      }
    }]
  })
});

// Liner WITHOUT it – the control
await fetch('/cart/add.js', {
  method: 'POST',
  headers: {'Content-Type': 'application/json'},
  body: JSON.stringify({items: [{id: LINER_VARIANT_ID, quantity: 1}]}
});

location.reload();
```

Theme path — if you want a real checkbox

Add to the cart line form in your cart template:

```
{%- if item.product.tags contains 'VAT Exempt Opt' -%}
<label>
  <input type="checkbox" name="properties[_vat_relief]" value="true"
    {% if item.properties['_vat_relief'] == 'true' %}checked{% endif %}>
  The price shown includes VAT. If you qualify for VAT relief, tick the box.
</label>
{%- endif -%}
```

Note: line properties can only be set when the item is added, so a cart-page checkbox has to re-add the line via `/cart/change.js`. The console method above sidesteps that for the purposes of this test.

2.4 What you should see

Where	Expected
Cart / checkout	Discount line “VAT relief” –£26.50
Subtotal	£191.00
Total	£164.50
VAT shown	£27.41
Liner	Untouched at £32.00 — no discount

2.5 Now open the order in admin — this is the actual test

Place the order. Admin → Orders → the order → look at the tax breakdown per line item.

Line	Charged	VAT recorded	Should be
AirTouch N30i (relieved)	132.50	22.08	0.00
REMZZZ liner	32.00	5.33	5.33
Order	164.50	27.41	5.33

The result: it fails, and it fails quietly

The buyer pays the right total. The checkout page looks correct. But Shopify recalculates tax on the *discounted* price, so the relieved line still carries **£22.08 of VAT**. ResMed remits that £22.08 out of margin on a supply HMRC says is zero-rated, and the invoice shows a VAT figure where it must show £0.00.

A discount reduces price. It cannot change taxability. That is the whole finding.

2.6 Prove it with the API rather than eyeballing the admin

```

query {
  order(id: "gid://shopify/Order/YOUR_ORDER_ID") {
    name
    totalPriceSet { shopMoney { amount } }
    totalTaxSet { shopMoney { amount } }
    lineItems(first: 10) {
      nodes {
        title
        originalTotalSet { shopMoney { amount } }
        discountedTotalSet { shopMoney { amount } }
        taxLines { title rate priceSet { shopMoney { amount } } }
        customAttributes { key value }
      }
    }
  }
}

```

The `taxLines` array on the relieved line item is the evidence. Screenshot it.

3. Test B — zero-rated variant swap

Same cart, different mechanism. Instead of discounting a taxable line, swap it for a variant that is genuinely non-taxable and priced ex-VAT.

Why the clone must be re-priced, not just marked non-taxable

Shopify Help Center, on tax-inclusive pricing: *“If you set a customer to be tax exempt, but you use tax-included pricing, then the customer is still charged the full listed product price.”*

So a non-taxable clone still priced at £159.00 means the buyer pays £159.00 and ResMed records £0.00 VAT — pocketing the VAT instead of passing the relief on. Worse than doing nothing, and it passes a careless “does the tax line say zero?” check. The clone must carry the net price, **gross ÷ 1.20 = £132.50**.

3.1 Two edits you have to make by hand

These were left out of the delivered archive so your `client_id` and any local changes were not overwritten.

```
# shopify.app.toml
scopes = "write_products,read_publications,write_publications"
```

```
// app/routes/app.tsx – inside <s-app-nav>
<s-link href="/app/vat-relief">VAT relief</s-link>
```

Then restart `shopify app dev` and re-approve the scope change when prompted.

3.2 Provision the zero-rated clones

1. **Open** the app → **VAT relief** (or go straight to `/app/vat-relief`).
2. **Create definitions**. Creates the two storefront-readable variant-pairing metafields. Safe to re-run.
3. **Read the Plan table**. For the AirTouch it must show gross `159.00` → clone `132.50` → VAT removed `26.50`. If it says anything else, stop — your prices or tax settings are not what you think they are.
4. **Provision clones**. Creates one hidden **draft** product per eligible product, variants non-taxable and net-priced, linked both ways by metafield.
5. **Publish** the clone product to the Online Store channel. It cannot be added to a cart while it is a draft.
6. **Check the Drift panel**. Should report no problems. It flags any clone that is still taxable or whose price has drifted off $\text{gross} \div 1.2$.

3.3 Install the theme side

1. Copy `theme/snippets/vat-relief-checkbox.liquid` and `theme/assets/vat-relief.js` into the theme.
2. Add to the cart template: `{{ 'vat-relief.js' | asset_url | script_tag }}`
3. In the cart line loop: `{% render 'vat-relief-checkbox', line_item: item, line_index: forloop.index %}`
4. Open `vat-relief.js` and set `CART_SECTION_ID` to your theme's cart section name.

3.4 Run it

1. Empty the cart. Add the AirTouch mask and the REMZZZ liner.
2. The checkbox should appear on the mask only. If it appears on the liner too, the eligibility tag is on the wrong product.
If it appears on neither, the metafield definitions are not storefront-readable — re-run step 2 of 3.2.
3. Tick it. The page reloads and the line is now the clone variant.

3.5 What you should see

Where	Expected
Cart line	Mask now £132.50, no discount line anywhere
Subtotal / total	£164.50
VAT shown	£5.33 — the liner only

Place the order and open the tax breakdown:

Line	Charged	VAT recorded	Verdict
AirTouch N30i (clone)	132.50	0.00	correct
REMZZZ liner	32.00	5.33	correct
Order	164.50	5.33	correct

Also confirm the order's line item still carries `_vat_relief: true` and `_vat_relief_declared_at` in its custom attributes. That is the HMRC self-declaration record, and it must survive onto the order.

4. Side by side — the result to show Betsy

Identical cart, identical buyer, identical total. Only the tax is different.

Same cart: AirTouch £159.00 (relief ticked) + REMZZZ liner £32.00	Test A discount	Test B variant swap
Buyer pays	164.50	164.50
VAT on the relieved line	22.08	0.00
VAT on the order	27.41	5.33
Invoice can show "VAT 20.0% £0.00" on the relieved line	no	yes
VAT ResMed remits on a zero-rated supply	22.08	0.00

Both approaches charge the buyer exactly the same £164.50. The difference is £22.08 per order that ResMed either remits to HMRC out of its own margin (Test A) or does not (Test B) — plus an invoice that is either wrong or right.

What each one costs

	Test A — discount	Test B — variant swap
Catalogue impact	none	one hidden product per eligible product
Inventory	none	duplicated; two variants cannot share stock
Price maintenance	none	clone must track gross ÷ 1.2 forever
Discount codes	stacks / conflicts with other discounts	clone excluded from product-scoped discounts
Reporting	clean	sales split across two product records
Tax correctness	wrong	right

The framing for the client conversation

This is not “Shopify can’t do it.” It is: the discount route keeps the catalogue clean and gets the tax wrong; the swap route gets the tax right and makes catalogue and inventory someone’s permanent job. That is a commercial decision about who absorbs which cost — not a technical dead end.

Why there is no third option

- **No tax Shopify Function exists.** The complete list is: cart & checkout validation, cart transform, delivery customization, discount, fulfillment constraints, order routing, payment customization, pickup generators.
- **Checkout UI extensions have no tax mutation.** The surface is `useApplyCartLinesChange`, `useApplyAttributeChange`, `useApplyDiscountCodeChange`.
- **Shopify’s Tax Platform** — the thing that could zero-rate a line properly — is invitation-only for tax software vendors. Not open to merchant apps.

- **Built-in exemption** is either customer-level (whole order) or product-level (always). Neither is per-line and per-buyer-declaration.

5. What to screenshot

1. Baseline checkout: £191.00 total, £31.83 VAT included.
2. Test A checkout: −£26.50 discount, £164.50 total. *This is the one that looks like success.*
3. **Test A order tax lines in admin: £22.08 on the relieved line.** This is the money shot.
4. Test B checkout: £164.50 total, £5.33 VAT.
5. **Test B order tax lines: £0.00 on the relieved line.**
6. The paper invoice sample next to both, with “VAT 20.0%” circled.

Show 2 before 3. Letting people see the plausible version first is what makes the failure land.

6. Troubleshooting

Symptom	Cause / fix
No discount appears in Test A	The automatic discount was created without <code>discountClasses: [PRODUCT]</code> , or the function returned early. Check the function run log in <code>shopify app dev</code> .
Discount applies to the liner too	The liner has picked up the <code>VAT Exempt Opt</code> tag, or the tag string does not match exactly — it is case and space sensitive.
Discount is £31.80 not £26.50	You are running a 20% version, not this one. 20% of a VAT-inclusive price is not the VAT.
Checkout extension missing from the editor	Dev store does not have Plus features enabled. Information / shipping / payment step extensions are Plus-only.
Checkbox does not render in Test B	Metafield definitions are not storefront-readable, or the variant has no pairing. Re-run “Create definitions” then “Provision clones”.
Clone cannot be added to cart	Still a draft. Publish it to the Online Store channel.
Clone charges £159.00 with £0.00 VAT	The clone was created taxable-false but not re-priced. The Drift panel will flag it. This is the failure mode called out in 3.
Totals drift after a price edit	Expected. Nothing syncs the pair automatically. Drift panel detects it; a <code>products/update</code> webhook would fix it properly.

7. Honest caveats

- **The Admin GraphQL in Test B is unverified at runtime.** It is written against the 2025-10 schema but has never been executed against a real store. Expect to correct a field name or two on first run — most likely in `productCreate`'s `productOptions` shape.
- **The clone is buyable by URL** once published. A determined buyer can order the net-priced variant without ever ticking the declaration. The line property is the compliance record, not the gate. Excluding the clone

tag from search and collections reduces the exposure; it does not close it.

- **The 20% rate is hardcoded** in both implementations. Fine for a test, not for production.
- **Neither test covers refunds or partial returns**, where the VAT treatment has to survive the reverse transaction. Worth raising before anyone commits to a route.
- **This guide is a test plan, not tax advice.** The HMRC position on zero-rating for chronically sick or disabled persons, and on what the invoice must show, needs confirming with ResMed's finance team before either approach goes near production.